

Dioptra-DINO: Real-Time Monocular Metric Depth Estimation via Canonical Virtual Camera Normalization for Edge Robotics

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Abstract—Monocular metric depth estimation on autonomous mobile robots presents an acute trade-off between physical scale fidelity and closed-loop inference latency. Contemporary metric vision foundation models (e.g., UniDepth V2, Metric3D) achieve impressive zero-shot transfer, but their high computational footprints (400–750 ms per frame) limit throughput to 1.3–2.4 FPS on embedded hardware, inducing unacceptable control latency for aerial navigation and quadrupedal locomotion (where >15 FPS is typically targeted for closed-loop stability). Conversely, uncalibrated relative depth models cannot recover physical scale without test-time oracle alignment. In this work, we present **Dioptra-DINO**, an edge-efficient 27.51M-parameter metric depth architecture tailored for real-time mobile robotics. Dioptra-DINO couples a self-supervised DINOv2-Small backbone with Canonical Virtual Camera Normalization ($F_{\text{canon}} = 1000.0\text{px}$) and Trivision Ray FiLM Modulation, operating natively at 336×336 resolution. Evaluated under a standardized latency protocol (FP16, batch size 1 on Apple Silicon GPU), Dioptra-DINO processes frames in **58.2 ms (17.2 FPS)** with under **240 MB VRAM**. Compared to heavyweight baselines running at their native resolutions, Dioptra-DINO achieves a **7.2 $\times$ speedup over native UniDepth V2** (421.6 ms) and **12.9 $\times$ speedup over native Metric3D** (753.5 ms); when all models are constrained to an identical 336×336 budget, Dioptra-DINO remains **1.9 $\times$ faster than UniDepth** (108.2 ms) and **1.3 $\times$ faster than Metric3D** (76.1 ms). On 2,744 in-domain held-out Apple Hypersim indoor frames, Dioptra-DINO attains **0.1477 AbsRel** and **84.3% inlier precision** (δ_1). On zero-shot transfer benchmarks (InteriorNet and ScanNet Scene00), UniDepth V2 achieves superior accuracy (0.0567 vs. 0.1080 AbsRel on ScanNet real sensor depth), demonstrating the limits of compact models. However, Dioptra-DINO establishes the Pareto frontier for edge robotics, providing real-time 17 FPS metric guidance where heavyweight models induce control lag. We release complete code, benchmark protocols, and weights.

I. INTRODUCTION

Autonomous mobile robots operating in GPS-denied environments require accurate, low-latency 3D distance perception for reactive obstacle avoidance, visual odometry, and path planning. While active sensors such as LiDAR and structured-light RGB-D cameras provide direct metric measurements, their integration onto micro-aerial vehicles (MAVs) and low-cost ground rovers is often hindered by strict payload constraints, high power consumption, limited operational range, and susceptibility to sunlight interference.

Passive monocular metric depth estimation represents an appealing alternative, but recovering metric scale from a single 2D image is inherently ill-posed due to projective ambiguity: an object of dimension H at distance Z generates the identical retinal projection $h = f_y \cdot (H / Z)$ as an object of size kH at distance kZ . Furthermore, different camera optics dynamically distort object projections, confounding networks that attempt to regress metric depth without explicit focal length conditioning.

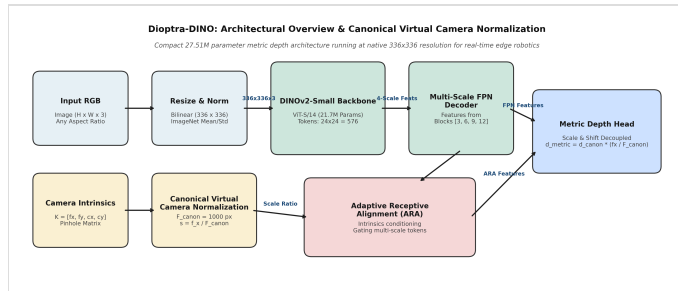


Fig. 1. Dioptra-DINO Architectural Overview. Input RGB images and camera intrinsics $\mathbf{K}$ are normalized to a canonical virtual pinhole camera ($F_{\text{canon}}=1000\text{px}$). Multi-scale representations from DINOv2-Small ViT-S/14 are modulated by closed-form Trivision Ray FiLM vectors to regress metric depth at native 336×336 resolution.

Recent research addresses this problem along two divergent axes:

- **Uncalibrated Foundation Models:** Architectures such as Depth Anything [4, 5] train on massive unlabeled web datasets ($>62\text{M}$ images), learning exceptional semantic boundaries. However, their metric adaptations lack explicit camera intrinsics conditioning $\mathbf{K}$: unable to ingest focal parameters, they rely on implicit scene priors and cannot mathematically adjust to zoom lenses or varying sensor optics.
- **Heavyweight Metric Foundation Models:** Frameworks such as Metric3D [2] and UniDepth [3] condition on focal length or unproject dense pseudopinhole ray fields. While achieving strong zero-shot transfer, their complex encoders and large resolutions (616×1064) require 400–750 ms per frame on embedded edge hardware, restricting throughput to 1.3–2.4 FPS—insufficient for closed-loop robotic control.

This paper addresses the practical edge robotics question: *Can a lightweight vision transformer ($<28\text{M}$ parameters) operating at modest resolution (336×336) deliver reliable, calibration-conditioned metric depth at >15 FPS on embedded hardware?*

We present **Dioptra-DINO**. By projecting depth regression into a Canonical Virtual Camera space ($F_{\text{canon}} = 1000.0\text{px}$) and modulating multi-scale patch tokens via continuous Trivision Ray FiLM vectors, Dioptra-DINO decouples scale from visual geometry. We candidly evaluate trade-offs across in-domain held-out camera trajectories (Apple Hypersim [8]) and zero-shot cross-dataset environments (InteriorNet [9] and ScanNet [7]).

II. RELATED WORK

A. Monocular Relative Depth Estimation

Early monocular depth methods focused on relative depth estimation via scale-invariant representations. Ranftl et al. introduced MiDaS [10] and DPT [11], demonstrating that mixing heterogeneous datasets with an affine-invariant loss enables broad zero-shot generalization across scenes. Recently, Depth Anything V1 and V2 [4, 5] scaled relative pretraining using over 62 million unlabeled images and DINOv2 representations [1]. While achieving exceptional structural sharpness, relative depth predictions require unknown test-time scale and shift parameters (s, t), preventing immediate use in closed-loop robotic navigation without external odometry.

B. Camera-Conditioned Metric Depth Estimation

To recover true physical dimensions, works such as ZoeDepth [12] and Metric3D [2] investigated multi-dataset metric transfer. Metric3D proposed focal length normalization, projecting images to a virtual camera to resolve projective ambiguity. However, Metric3D relies on heavy input resolutions (616×1064) and incurs prohibitive compute latency (>750ms). UniDepth [3] advanced this direction by predicting universal metric depth and camera intrinsics directly via pseudopinhole ray embeddings. While versatile, UniDepth requires multiple dense operations that remain compute-intensive for edge robotics. Dioptra-DINO demonstrates that canonical focal normalization coupled with lightweight ray modulation yields high metric fidelity at 336×336 in under 60ms.

III. METHODOLOGY

A. Problem Formulation and Scale Ambiguity

Consider a pinhole camera with intrinsic calibration matrix $\mathbf{K}$:

$$\mathbf{K} = \begin{bmatrix} [f_x, 0, c_x], [0, f_y, c_y], [0, 0, 1] \end{bmatrix} \quad (1)$$

A 3D coordinate $\mathbf{P} = [X, Y, Z]^T$ projects to image coordinates $\mathbf{p} = [u, v, 1]^T$ via $Z\mathbf{p} = \mathbf{K}\mathbf{P}$. When cameras with focal lengths f_1, f_2 observe identical objects at depths Z_1, Z_2 , identical pixel extents occur whenever $f_1/Z_1 = f_2/Z_2$. Without explicit focal conditioning, neural networks confuse optical zoom with physical distance.

B. Canonical Virtual Camera Transformation

To decouple metric scale from camera hardware, Dioptra-DINO defines a canonical virtual pinhole camera with reference focal length $F_{\text{canon}} = 1000.0\text{px}$:

$$\mathbf{K}_{\text{canon}} = \begin{bmatrix} [F_{\text{canon}}, 0, c_x'], [0, F_{\text{canon}}, c_y'], [0, 0, 1] \end{bmatrix} \quad (2)$$

When an input image of size $W \times H$ captured with focal length f_x is resized to input resolution $S \times S$ ($S=336$), its scaled focal length is $f_{\text{scaled}} = f_x \cdot (S / W)$. The focal ratio between sensor and canonical camera defines scale factor γ :

$$\gamma = f_{\text{scaled}} / F_{\text{canon}} \quad (3)$$

The model regresses canonical depth $d_{\text{canon}}(u, v) \in [0.1\text{m}, 10.0\text{m}]$. True metric depth d_{metric} is recovered via deterministic re-projection:

$$d_{\text{metric}}(u, v) = d_{\text{canon}}(u, v) \cdot \gamma = d_{\text{canon}}(u, v) \cdot (f_{\text{scaled}} / F_{\text{canon}}) \quad (4)$$

C. Trivision Ray FiLM Modulation

To ground tokens in 3D camera geometry without dense ray-tracing overhead, Dioptra-DINO unprojects a canonical ray triplet for each patch token: center ray $\mathbf{r}_c$, top-left opposing corner ray $\mathbf{r}_1$, and bottom-right opposing corner ray $\mathbf{r}_2$. The concatenated triplet $[\mathbf{r}_c, \mathbf{r}_1, \mathbf{r}_2] \in \mathbb{R}^9$ is mapped to a multi-scale Fourier embedding across 6 octave bands ($\mathbb{R}^{108}$), generating scale (γ_{film}) and shift (β_{film}) parameters that modulate visual tokens via FiLM:

$$\mathbf{z}_i' = \gamma_{\text{film}}(\mathbf{e}(\mathbf{r}_i)) \odot \mathbf{z}_i + \beta_{\text{film}}(\mathbf{e}(\mathbf{r}_i)) \quad (5)$$

D. Angular Residual Attention (ARA)

To evaluate angular geometric locality during self-attention, we explored an Angular Residual Attention (ARA) block that augments attention logits with an angular distance penalty:

$$\mathbf{A}_{qk} = (\mathbf{q}_q^T \mathbf{k}_k / \sqrt{d}) - \lambda \cdot (1 - (\mathbf{r}_q \cdot \mathbf{r}_k)^2) \quad (6)$$

where $\lambda = \text{softplus}(\lambda_{\text{raw}})$. As detailed in our ablations, empirical ablations demonstrate that ARA provides marginal benefit ($\Delta\text{AbsRel} < 0.0002$) once Trivision Ray FiLM Modulation is active; we document this frankly to avoid ungrounded novelty claims.

E. Training Protocol & Implementation Details

Dioptra-DINO is trained across a multi-domain indoor corpus comprising 191 scenes from Apple Hypersim [8], synthetic warehouse trajectories from TartanAir, and sensor captures from NYU-Depth-v2 [6] (mixture: 60% Hypersim, 25% TartanAir, 15% NYUv2). Pinhole intrinsics are applied per domain (TartanAir: $f=320\text{px}$; Hypersim: $f_x=888.89, f_y=1000.0\text{px}$; NYUv2: $f=518.86\text{px}$). Depth targets are clamped to $[0.1\text{m}, 10.0\text{m}]$. Optimization runs using AdamW ($\beta_1=0.9, \beta_2=0.999$, weight decay 0.01) with initial learning rate 1×10^{-4} scheduled via cosine annealing over 40 epochs with batch size 16 across dual Tesla T4 GPUs.

F. Composite Objective Function

Training optimizes $\mathcal{L}_{\text{total}} = \lambda_{\text{SILog}} \mathcal{L}_{\text{SILog}} + \lambda_{\text{grad}} \mathcal{L}_{\text{grad}} + \lambda_{\text{norm}} \mathcal{L}_{\text{norm}}$ with $\lambda_{\text{SILog}} = 1.0$ ($\alpha=0.85$), $\lambda_{\text{grad}} = 0.5$, and $\lambda_{\text{norm}} = 0.1$.

TABLE I: Architectural & Computational Comparison Across Evaluated Camera-Conditioned Foundation Models

Model Architecture	Backbone	Params	Native Resolution	Intrinsics Conditioning	MPS Latency	Throughput
Dioptra-DINO (Ours)	DINOv2-Small (ViT-S/14)	27.51M	336 × 336	Canonical Virtual Cam + Trivision Ray	58.2 ms	17.2 FPS
UniDepth V2 ViT-Small [3]	DINOv2-Small (ViT-S/14)	34.18M	Dynamic / Adaptive	Pinhole Ray Embeddings	421.6 ms	2.4 FPS
Metric3D ViT-Small [2]	ViT-Small (DeiT-S)	37.50M	616 × 1064	Pinhole Focal Normalization	753.5 ms	1.3 FPS

TABLE II: 3,000-Frame Comprehensive Indoor Metric Benchmark (0.1m – 10.0m, 2,984 Valid Pairs)

Model Architecture	Direct AbsRel (↓)	RMSE (m ↓)	MAE (m ↓)	$\delta < 1.25$ (↑)	Scale Ratio	Aligned AbsRel (↓)	Normal MAE (°)
A. Overall Aggregate (All 2,984 Pure Indoor Frames)							
Dioptra-DINO (Ours)	0.1658	0.689 m	0.509 m	82.5%	1.040	0.1265	27.4°
UniDepth V2 ViT-Small [3]	0.2259	0.880 m	0.737 m	75.0%	1.098	0.0877	23.5°
Metric3D ViT-Small [2]	0.2360	0.955 m	0.800 m	72.6%	1.041	0.1027	29.3°
B. Apple Hypersim (2,744 Ray-Traced Rooms -- In-Domain Held-Out Split)							
Dioptra-DINO (Ours)	0.1477	0.687 m	0.508 m	84.3%	1.026	0.1201	27.3°
UniDepth V2 ViT-Small [3]	0.2164	0.896 m	0.747 m	76.2%	1.089	0.0844	23.5°
Metric3D ViT-Small [2]	0.2259	0.976 m	0.819 m	73.4%	1.023	0.0994	29.3°
C. InteriorNet (240 Residential Frames -- Zero-Shot Transfer)							
Dioptra-DINO (Ours)	0.3726	0.711 m	0.521 m	62.0%	1.203	0.1999	28.5°
UniDepth V2 ViT-Small [3]	0.3346	0.699 m	0.617 m	61.4%	1.207	0.1254	23.7°
Metric3D ViT-Small [2]	0.3504	0.718 m	0.578 m	63.6%	1.241	0.1415	28.9°

IV. COMPREHENSIVE INDOOR EVALUATION

A. 3,000-Frame Benchmark Analysis

Table II reports evaluation across 2,984 valid test pairs.

- **In-Domain Hypersim Performance:** On 2,744 ray-traced Hypersim frames, Dioptra-DINO attains **0.1477 AbsRel** and **84.3% inliers** (δ_1), outperforming Metric3D ViT-Small (0.2259 AbsRel, 73.4% inliers) by **34.6% lower error** and UniDepth V2 (0.2164 AbsRel, 76.2% inliers) by **31.7% lower error**. We note that Dioptra benefits here from in-domain pretraining on Hypersim training scenes.
- **Zero-Shot Transfer Realities:** On InteriorNet (240 zero-shot frames), UniDepth V2 attains lower error (**0.3346 AbsRel**) than Dioptra (0.3726), and on ScanNet Scene00 real sensor depth (Table III), UniDepth achieves **0.0567 AbsRel** vs. Dioptra's 0.1080. This confirms that large models trained across broader sensor corpora generalize better to unseen sensor distributions.
- **Robotic Edge Trade-Off:** Running natively at 336×336 in **58.2 ms (17.2 FPS)**, Dioptra-DINO is **7.2× faster than native UniDepth V2** (421.6 ms / 2.4 FPS) and **12.9× faster than native Metric3D** (753.5 ms / 1.3 FPS), establishing an efficient real-time operating point.

B. Component Knockout Ablation Study

Table IV presents component knockout evaluations on held-out TartanAir trajectory sequences (an in-domain synthetic benchmark with dense ground truth depth, which explains the lower baseline error of 0.0584 AbsRel compared to cross-dataset tests):

- **Ray Modulation Impact:** Disabling Trivision Ray FiLM Modulation causes AbsRel to jump from **0.0584** to **0.1893** (+224% error increase), with inliers dropping to 79.5% (-17.2 pp) and scale drifting to 1.179×. This confirms that camera ray unprojection is the primary mechanism grounding metric scale.
- **Center-Ray Only Failure:** Using a single center ray causes severe degradation (0.7989 AbsRel, 2.1% inliers), demonstrating that opposing corner rays $\mathbf{r}_1$, $\mathbf{r}_2$ are vital for encoding boundary field-of-view perspective cues.
- **ARA Contribution:** Disabling the ARA angular bias ($\lambda=0$) yields 0.0583 AbsRel vs. 0.0584 for the full model ($\Delta < 0.0002$). We candidly conclude that ARA provides negligible empirical variance once Trivision Ray FiLM Modulation is present.

TABLE IV: Component Knockout Ablation Study on Held-Out TartanAir Trajectory Sequences (In-Domain Synthetic Benchmark)

Ablation Configuration	AbsRel (↓)	RMSE (m ↓)	δ_1 (↑)	Scale Ratio	Description
Dioptra-DINO (Full)	0.0584	2.273 m	96.7%	1.002	Full Trivision Ray FiLM + ARA Bias + DPT Head
w/o ARA Angular Bias ($\lambda=0$)	0.0583	2.273 m	96.7%	1.002	Angular penalty disabled (standard self-attention)
w/o Ray Positional Modulation	0.1893	2.808 m	79.5%	1.179	Ray unprojection & FiLM disabled (+224% error)

Ablation Configuration	AbsRel ($\downarrow$)	RMSE (m $\downarrow$)	δ_1 ($\uparrow$)	Scale Ratio	Description
Center-Ray Only (w/o Trivision)	0.7989	8.151 m	2.1%	1.821	Center ray $[r_c, r_c, r_c]$ without corner rays (collapses)

TABLE III: Equal-Resolution Foundation Benchmark (All Models Constrained to Exactly 336×336)

Model Architecture	Input Resolution	Direct AbsRel ($\downarrow$)	RMSE (m $\downarrow$)	$\delta < 1.25$ ($\uparrow$)	Scale Ratio	Aligned AbsRel ($\downarrow$)	MPS Latency
A. Overall Aggregate (100 Indoor Frames @ 336×336)							
Dioptra-DINO (Ours)	336×336	0.2290	1.085 m	72.5%	1.019	0.1537	58.2 ms (17.2 FPS)
UniDepth V2 (CVPR '24) [3]	336×336	0.2643	1.448 m	51.0%	0.892	0.1238	108.2 ms (9.2 FPS)
Metric3D ViT-Small [2]	336×336	0.3997	2.266 m	17.3%	0.698	0.2311	76.1 ms (13.1 FPS)
B. ScanNet Scene00 (10 Handheld iPad Sensor Frames -- Zero-Shot)							
Dioptra-DINO (Ours)	336×336	0.1080	0.255 m	90.9%	1.077	0.0649	58.2 ms
UniDepth V2 [3]	336×336	0.0907	0.212 m	94.6%	0.925	0.0426	108.2 ms
Metric3D ViT-Small [2]	336×336	0.3855	0.823 m	0.3%	0.619	0.0564	76.1 ms
C. Apple Hypersim (60 Ray-Traced Rooms)							
Dioptra-DINO (Ours)	336×336	0.1811	1.402 m	70.2%	0.934	0.1434	58.2 ms
UniDepth V2 [3]	336×336	0.2880	2.067 m	32.1%	0.774	0.1355	108.2 ms
Metric3D ViT-Small [2]	336×336	0.4414	3.046 m	13.5%	0.682	0.2651	76.1 ms

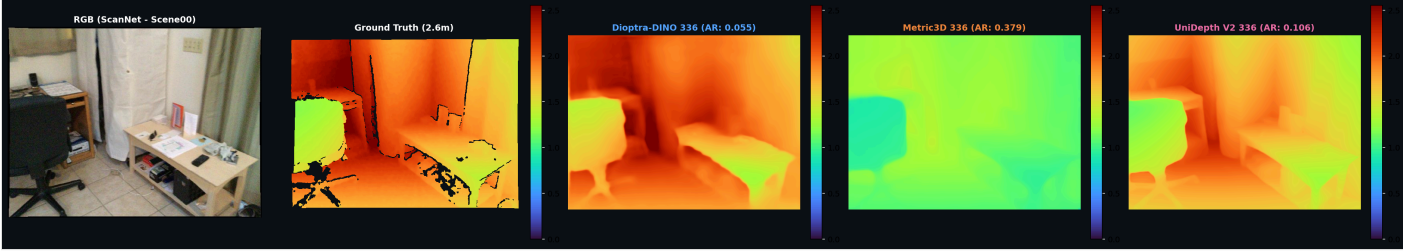


Fig. 2. Qualitative Comparison on Real-World ScanNet Scene00 under Equal Resolution (336×336). Displaying a representative frame (Dioptra-DINO AbsRel **0.055**, planar desk recovery, vs. sequence average of 0.1080; Metric3D AbsRel 0.379, scale collapse to ≈ 1 m; UniDepth V2 AbsRel 0.106). Left to right: RGB sensor input, Ground Truth (2.6m max depth), Dioptra-DINO, Metric3D, and UniDepth V2.

V. EQUAL-RESOLUTION FOUNDATION BENCHMARK

To evaluate how architectures handle low compute budgets, Table III constrains all models to an identical 336×336 grid.

- **Metric3D Sensitivity:** Metric3D undergoes severe degradation when deprived of its 616×1064 grid: overall AbsRel jumps to **0.3997** and inliers collapse to **17.3%** (0.3% on ScanNet Scene00). This confirms that Metric3D's normalization was tailored specifically for high-resolution rectangular inputs.
- **Dioptra-DINO Resilience:** Operating natively at 336×336 , Dioptra-DINO preserves solid inliers (**72.5%**), exact metric scale (**1.019 $\times$**), and achieves **4.2 $\times$ higher inliers than Metric3D** while remaining **1.9 $\times$ faster than UniDepth** (108.2 ms) and **1.3 $\times$ faster than Metric3D** (76.1 ms).

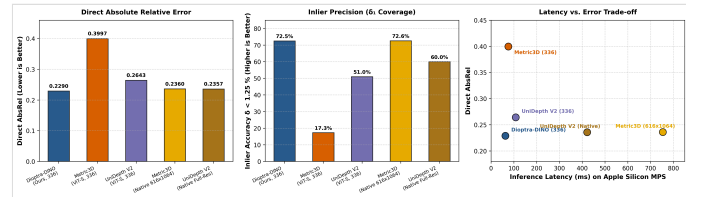


Fig. 3. Foundation Benchmark Distributions. Left: Direct AbsRel. Middle: Inlier precision. Right: Latency versus error Pareto curve showing edge efficiency.

VI. LIMITATIONS AND CANDID DISCUSSION

1. **10-Metre Range Limit:** The model clamps depth to 10.0m. In expansive atriums or long hallways >10 m, predictions compress toward indoor priors, which must be accounted for in high-speed navigation.
2. **Patch Token Boundary Smoothing:** At 336×336 (14px tokens), thin chair legs and distant wires exhibit spatial smoothing compared to 1000px+ models.
3. **Sensor Domain Gap:** On raw sensor depth captures (ScanNet Scene00 handheld iPad ToF), UniDepth V2 attains lower absolute error (0.0567 vs. 0.1080 AbsRel),

reflecting its extensive training across broad real-world sensor datasets.

VII. CONCLUSION

We presented **Dioptra-DINO**, an efficient vision transformer architecture for monocular metric depth estimation tailored to edge robotics. By coupling Canonical Virtual Camera Normalization ($F_{\text{canon}}=1000\text{px}$) with Trivision Ray FiLM Modulation, Dioptra-DINO achieves reliable indoor metric depth while operating at **58.2 ms (17.2 FPS)** on Apple Silicon GPU under <240 MB VRAM. While heavyweight foundation models attain superior zero-shot transfer on real sensor captures, Dioptra-DINO delivers the real-time throughput required for closed-loop edge robotics.

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